

A Multi-Equation LIVES System for Australia: Joint Identification of Credit Conditions and Williams' Calibrations Tested

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1 A Multi-Equation LIVES System for Australia: Joint Identification of Credit Conditions and Williams' Calibrations Tested

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1.1 Abstract

We extend the single-equation Australian LIVES consumption model in our headline paper to a multi-equation system, following Williams' (2010, 2012) four-equation specification: consumption, house prices, mortgage stock, and home equity withdrawal jointly estimated under cross-equation sign restrictions. We implement the system in four sequential stages: a two-equation SUR of consumption and house prices, a three-equation joint sign-survival identification of the credit-conditions index (CCI), a four-equation system that adds an HEW equation, and a formal Wald test of Williams' Table 1 calibrations as restrictions on the consumption equation.

The empirical results are sharper than the single-equation evidence in the headline paper. **Joint sign-survival of CCI knots collapses dramatically as more equations are added.** From the 5 knots that survive the iterated consumption-equation maximal-GETS reduction (2009Q1, 2019Q1, 2020Q2, and two others), only 2 survive a three-equation joint test (1986 financial deregulation; 2017 APRA macroprudential round II), and only 1 survives the four-equation test once the HEW equation is added (1986 deregulation alone). **Williams' $\zeta_h = 1$ normalisation becomes inert with one surviving knot** — the HP-weighted, M-weighted, and consumption-weighted CCI variants are mathematically identical after peak-normalisation, so the cross-equation rescaling has no traction.

Williams' Table 1 wealth γ are formally rejected on the contemporary Australian data. A joint Wald test using the Newey–West covariance gives $\chi^2(4) = 10.03$ ($p = 0.040$) for the four wealth calibrations alone and $\chi^2(6) = 29.10$ ($p < 0.001$) for the full Williams calibration set. The individual rejection is the housing-wealth m.p.c. $\gamma_{HA} = 0.0488$ ($\chi^2(1) = 7.18$, $p = 0.007$), even though the *implied* structural γ_{HA} from our Spec 6 estimate is 0.049 — essentially equal to Williams' value. The reconciliation runs through λ : the OLS coefficient on ha_y (0.0088) sits 37 per cent below Williams' implied OLS at his published λ (0.0140), and that gap is what generates the rejection of the calibration as a parameter restriction.

The HEW equation we construct using $\Delta(\text{housing-secured debt}) / \text{income}$ as a proxy (Williams' literal definition is unsourced for contemporary Australia) yields a residual correlation of 0.83 with the mortgage-stock equation — confirming that under the proxy construction the two equations are not separately identifying signals. A more discriminating HEW measure, with the dwelling-investment subtraction Williams' definition includes, is the binding constraint on further multi-equation work.

The Phase A multi-equation system therefore complements rather than overturns the headline paper's positive finding: Spec 6 reproduces Williams' structural γ profile to a useful approximation, and the multi-equation extension formally tests *and rejects* the calibrated values as a system of parameter restrictions. The four-equation common-factor hypothesis — that a single CCI series enters all four LIVES equations with sign-consistent loadings — is only weakly supported on contemporary Australian data, with one institutional event (1986 deregulation) carrying the joint identification.

1.2 1. Introduction

This paper is the companion to our headline single-equation paper on the Australian LIVES consumption model. The headline paper estimates Williams (2010, 2012) consumption equation on contemporary public data and finds a structural γ profile broadly consistent with Williams' Table 1: $\gamma_{HA} = 0.049$ (Williams: 0.049), $\gamma_{IFA} = 0.030$ (Williams: 0.022), $\gamma_{NLA} = 0.196$ (Williams: 0.159). The single-equation framing is positive about Williams' calibration set in aggregate, but cannot impose the cross-equation parameter restrictions that the LIVES framework treats as the structural identification of the credit-conditions index.

This paper builds the multi-equation system in stages. The objective is to assess whether the cross-equation parameter restrictions in the four-equation LIVES system are empirically supported on contemporary Australian data, and whether they close the residual identification questions left open by the single-equation paper.

1.2.1 1.1 Headline findings

1. **Cross-equation joint sign-survival of the CCI knots collapses as equations are added.** Three knots survive iterated consumption-equation fitting (2009Q1, 2019Q1, 2020Q2); two survive the three-equation joint test (1986, 2017); one survives the four-equation test once the HEW equation is added (1986 alone).
2. **$\zeta_h = 1$ normalisation has no empirical traction with one surviving knot.** Williams' identification scheme — pinning the HP-equation CCI loading to 1 and estimating $\zeta_c, \zeta_m, \zeta_w$ as relative scalings — requires multiple surviving knots whose relative weights can shift across normalisation schemes. With one surviving knot the three weighted variants are identical after peak-normalisation.
3. **Williams' Table 1 calibrations are formally rejected.** A Wald test of the joint $\gamma_{HA} = 0.0488$, $\gamma_{IFA} = 0.022$, $\gamma_{NLA} = 0.159$, $\psi_0 = 0.20$, and $\ln(p^h/y) = -0.13$ restrictions gives $\chi^2(6) = 29.10$, $p < 0.001$. The wealth-only subset rejects at 5 per cent, $\chi^2(4) = 10.03$, $p = 0.040$. The individual rejection is γ_{HA} — and yet the *implied* γ_{HA} from our Spec 6 estimate (0.049) is essentially equal to Williams' value, so the rejection runs through the speed-of-adjustment channel rather than the wealth-coefficient channel.
4. **The HEW residuals correlate 0.83 with mortgage-stock residuals** under the credit-flow proxy construction. The two equations are not separately identifying signals; a properly constructed HEW series with the dwelling-investment subtraction Williams' definition includes would be the next data investment.

1.2.2 1.2 Relationship to the headline paper

The headline single-equation paper finds that Spec 6 (the preferred disaggregated specification with CCI short-run dynamics and the post-2008 PI break) reproduces Williams' Table 1 γ profile to a useful approximation. That finding stands: the present multi-equation paper does not overturn it. But the multi-equation analysis here clarifies *why* the positive single-equation reading coexists with the formal rejection of Williams' calibrations as parameter restrictions — the headline paper's γ are implied quantities from the ECM identity, while the calibration restrictions test the OLS coefficients directly, and our smaller $|\lambda|$ breaks the correspondence between the two test forms.

1.3 2. Methodology

1.3.1 2.1 The four-equation LIVES system

Williams (2010) estimates four equations jointly by full-information maximum likelihood:

$$\begin{aligned}\Delta \log c_t &= \theta_C \left[\dots + \gamma_{HA} \cdot (HA/y) \cdot (1 - \varpi \cdot CCI) + \psi_t \cdot \log(\hat{y}^p/y) \right. \\ &\quad \left. + \zeta_c \cdot CCI + \lambda_C \cdot ecm_lag_C \right] + \sum \beta_C \cdot SR_C + \varepsilon_{C,t} \\ \Delta \log hp_t &= \theta_H \left[\dots + \alpha_R \cdot r \cdot (1 + \kappa \cdot CCI) + \zeta_h \cdot CCI + \lambda_H \cdot ecm_lag_H \right] \\ &\quad + \sum \beta_H \cdot SR_H + \varepsilon_{H,t} \\ \Delta \log m_t &= \theta_M \left[\dots + \eta \cdot CCI + \lambda_M \cdot ecm_lag_M \right] + \sum \beta_M \cdot SR_M + \varepsilon_{M,t} \\ hew_t &= \theta_W \left[\dots + (HLI \text{ factor}) \cdot CCI + \lambda_W \cdot ecm_lag_W \right] \cdot 1/(HA/y) \\ &\quad + \sum \beta_W \cdot SR_W + \varepsilon_{W,t}\end{aligned}$$

with $\zeta_h = 1$ as identification, $\zeta_c, \zeta_m, \zeta_w$ as relative scalings, and ϖ restricted to the same value across the consumption and HP equations.

1.3.2 2.2 Joint sign-survival as a tractable approximation to FIML

We do not implement full FIML in this paper. The structural restriction at the heart of LIVES — that the *same* CCI series enters all four equations with sign-consistent loadings — can be tested approximately via **joint sign-survival**: fit each equation separately with the full 15-knot smoothed-step CCI basis, and retain only knots whose coefficient sign matches its institutional prior in every equation simultaneously. The retained set defines the joint-identified CCI; the relative loadings on this CCI across equations are the empirical analogues of Williams' ζ_i .

This approximation imposes a sign restriction, not a parameter equality restriction. True FIML imposes both. We document where the two differ.

1.3.3 2.3 $\zeta_h = 1$ normalisation

After joint sign-survival, we construct three CCI variants by weighting the surviving knots with each equation's coefficients:

- `cci_williams_joint` — consumption-equation weights
- `cci_williams_joint_h` — house-price-equation weights (Williams' $\zeta_h = 1$)
- `cci_williams_joint_m` — mortgage-equation weights

Each variant is peak-normalised. The HP-weighted variant is the literal implementation of Williams' $\zeta_h = 1$ normalisation: the CCI is the linear combination of surviving knots whose loading in the HP equation is unity by construction. The other equations' loadings on this rescaled series are the relative scalings $\zeta_c, \zeta_m, \zeta_w$. This identification matters when

multiple knots survive: the relative weights then differ across the variants and the CCI shapes are distinct. With only one surviving knot the variants collapse to the same series.

1.3.4 2.4 Williams' calibrations as Wald restrictions

Williams reports six calibrated long-run γ in Table 1: $\gamma_{HA} \in [0.0452, 0.0488]$ (housing wealth at CCI peak), $\gamma_{IFA} = 0.022$ (illiquid financial, calibrated), $\gamma_{NLA} = 0.159$ (net liquid), $\psi_0 = 0.20$ (PI weight at CCI = 0), the wealth-CCI interaction $\omega = 1.2$, and the speed of adjustment $\lambda = -0.286$.

The ECM identity $OLS_coef_i = \lambda \times \gamma_i$ lets us cast each calibration as a linear restriction on the OLS coefficient vector of our Spec 6, conditional on λ . We test these restrictions individually, in groups, and jointly using `car::linearHypothesis` with the Newey–West covariance.

1.4 3. Data and equations

We use the back-extended master dataset from the headline paper (1976Q3–2024Q4, $n = 194$ quarters; see §3 of the headline paper for construction). The four equations are estimated on the common sample where all four LHS variables and all RHS regressors are non-missing.

1.4.1 3.1 The HEW proxy

Williams (Aust paper §4.4) defines home equity withdrawal as

$$HEW_t = \Delta(\text{housing-secured debt})_t + \text{housing-related grants}_t - \text{dwelling investment}_t$$

The RBA publishes the constructed HEW series as an unpublished internal time series in Williams' work; it is not available in the current public data vintage. We construct a proxy that uses the credit-flow dimension only:

$$hew_proxy_t = \Delta(\text{fin_loans_proxy})_t / ydi_ann_nom_t$$

with two caveats: (i) the dwelling-investment subtraction is omitted, so our proxy over-states "true" HEW; (ii) `fin_loans_proxy` is total household debt rather than housing-specific debt — on the modern sample housing-specific debt is ~85 per cent of the total, so the approximation is close, but the pre-1990 portion of the sample is noisier.

The proxy is sufficient to demonstrate the multi-equation framework's mechanics; quantitatively reliable HEW estimates would require sourcing the RBA's unpublished series or constructing a properly deflated dwelling-investment time series.

1.4.2 3.2 Equations estimated

Eq	Response	Long-run regressors			ECM term
C	$\Delta \log c$	$\ln_networth_y$,	$\ln_hp_over_y$,	$real_rate$,	$\log(c_{t-1}/y_t)$
		$\ln_yp_over_y$, CCI			
H	$\Delta \log hp_real$	$lincome$, $\log_credit_y$,	$real_rate$, $prime_age_share$,		$\log(hp_{t-1})$
		CCI			
M	$\Delta \log M_real$	$lincome$,	$\ln_hp_over_yd$,	$real_rate$,	$\log(M_{t-1})$
		$prime_age_share$, CCI			
W	hew_proxy	$lincome$, ha_y_proxy ,	$real_rate$, $prime_age_share$,		hew_proxy_{t-1}
		CCI			

Sign priors on the CCI loading: positive in all four equations (looser credit conditions raise consumption, house prices, mortgage stock, and home-equity withdrawal).

1.5 4. Phase 1 — two-equation SUR

The two-equation SUR of consumption and house prices on the back- extended sample ($n = 189$, 1977Q4–2024Q4) is the foundation result. The residual correlation between the two equations is $\hat{\rho}(\epsilon_C, \epsilon_H) = 0.0007$ — essentially zero — under the full specification with `cci_williams` and event dummies. The finding is robust across variants: removing CCI gives $\hat{\rho} = -0.083$, removing event dummies gives $\hat{\rho} = +0.043$, and the minimal LR + SR specification gives $\hat{\rho} = -0.025$.

Joint estimation therefore delivers no efficiency gain at the quarterly frequency. The case for the multi-equation framework rests entirely on cross-equation parameter restrictions, not on residual covariance.

A sign violation is documented on `cci_williams` in the house-price equation: loading -0.024 , $t = -4.68$, against an institutional prior of positive. This is the structural failure that motivates the joint-identified CCI of phases 3 and beyond.

1.6 5. Phase 3 — three-equation joint sign-survival

We refit all three equations (consumption, HP, mortgage stock) with the full 15-knot Williams CCI basis on the back-extended common sample ($n = 189$). A knot is **jointly surviving** if its coefficient sign matches the institutional prior in every equation simultaneously.

1.6.1 5.1 Survival results

Survival regime	Surviving knots	n
Consumption only (iterated)	1979Q1, 1986Q1, 1992Q1, 2007Q3, 2017Q1, 2020Q2	6
Three-equation joint (C $\cap$ H $\cap$ M)	1986Q1, 2017Q1	2

Four of the six consumption-survivors fail joint sign tests in the HP or mortgage equations. The retained set — 1986 financial deregulation and 2017 APRA macroprudential round II — covers two broad-spectrum institutional events with sectoral effects on all three blocks.

1.6.2 5.2 Joint identification fixes the HP sign violation

Constructing `cci_williams_joint` from the two surviving knots, weighted by consumption-equation coefficients, and re-running the HP equation:

HP-equation CCI loading	(a) cons-only CCI	(b) joint CCI
Estimate	−0.024	+0.024
t-statistic	−4.68	+2.03
Sign vs prior (+)	× violator	✓

Joint identification flips the HP equation's CCI sign from significantly wrong to significantly right. This is Williams' cross-equation sign restriction working as intended.

1.6.3 5.3 Joint identification does NOT fix the M sign violation

The mortgage equation's CCI loading remains wrong-signed (loading −0.0086, $t = -1.43$) under the joint-identified CCI. The two-knot weighted construction is insufficient to pass sign tests in all three equations simultaneously when the weights derive from one equation's coefficients. True FIML imposes parameter equality (magnitude as well as sign) across equations; the joint sign-survival approximation imposes only sign equality.

1.7 6. Phase A — four-equation system with HEW

1.7.1 6.1 Adding the HEW equation collapses the joint survivor set

Refitting all four equations (consumption + HP + M + HEW) on the common sample with the 15-knot CCI basis, and requiring sign-prior survival in every equation, leaves **only one knot**:

Survival regime	Surviving knots	n
Three-equation joint (C ∩ H ∩ M)	1986Q1, 2017Q1	2
Four-equation joint (C ∩ H ∩ M ∩ W)	1986Q1	1

`Sdmma_2017_03` sign-violates in the HEW equation (HEW responds positively to APRA tightening in the data, against the institutional prior of negative). Only the 1986 financial deregulation knot survives the four-equation joint test.

1.7.2 6.2 $\zeta_h = 1$ normalisation is inert with one surviving knot

With one surviving knot, the three weighted variants of `cci_williams_joint` (consumption-weighted, HP-weighted, mortgage-weighted) are mathematically identical after peak-normalisation. The variant *correlations are all 1.0*. The HP-equation CCI loading is identical across all three regimes, as is the mortgage-equation loading and the HEW-equation loading. The cross-equation rescaling of Williams' $\zeta_h = 1$ normalisation has no empirical traction.

This is not a coding artefact: it is a property of the underlying data. With one surviving knot, the CCI is essentially a single institutional shock (the 1986 deregulation smoothed-step). Any positive multiplicative rescaling of this shock leaves its peak-normalised form unchanged.

1.7.3 6.3 HEW residuals correlate 0.83 with mortgage residuals

The four-equation SUR residual correlation matrix (under the joint-identified CCI):

	C	H	M	W
C	1.000	-0.107	+0.090	+0.042
H	-0.107	1.000	-0.194	-0.183
M	+0.090	-0.194	1.000	+0.832
W	+0.042	-0.183	+0.832	1.000

The mortgage and HEW residuals correlate at **+0.83**. This is a structural feature of the proxy HEW construction: $\text{hew_proxy} = \Delta \text{fin_loans_proxy} / \text{income}$ is the change-form of the M-equation dependent variable $\log M_{\text{real}}$. Under the proxy the two equations are not separately identifying signals.

1.7.4 6.4 Coefficient profile across regimes

The 4-equation SUR coefficient profile is summarised in `lives_phase_a_summary.csv`. Headline λ values across the four equations under the joint CCI regime:

Equation	λ_{eq}
Consumption	-0.215
House price	-0.037
Mortgage	+0.002 (essentially zero)
HEW	+0.067 (wrong sign)

The HEW equation's ECM term is wrong-signed under the proxy construction — hew_proxy does not mean-revert in our sample. A properly constructed HEW measure with the dwelling-investment subtraction would likely behave differently here.

1.8 7. Phase B — Williams' Table 1 calibrations as Wald restrictions

We refit Spec 6 (the preferred disaggregated specification from the headline paper) on the canonical Italy-LP master and test Williams' Table 1 calibrations as linear restrictions on the OLS coefficient vector, conditional on the estimated $\tilde{\lambda}$.

1.8.1 7.1 Calibration grid

Coefficient	Williams γ	Implied OLS at $\tilde{\lambda}$
ha_y	0.0488	-0.0096
eq_y	0.0110	-0.0022
super_y	0.0110	-0.0022
nla_y	0.1590	-0.0314
ln_hp_over_y	-0.1300	+0.0257
ln_yp_over_y	0.2000	-0.0395

The implied OLS values are computed at $\tilde{\lambda} = -0.197$ (the version of Spec 6 estimated by the test script; this is a slight refit of the canonical -0.180 in the headline paper because the script re-derives the data flow rather than reusing the cached pipeline outputs).

1.8.2 7.2 Wald test results

Restriction	χ^2	df	p-value	Reject at 5 %	Reject at 1 %
ha_y = 0.0488	7.18	1	0.007	✓	✓
eq_y = 0.011	0.02	1	0.896	×	×
super_y = 0.011	2.26	1	0.133	×	×
nla_y = 0.159	2.55	1	0.110	×	×
ln_hp_over_y = -0.130	2.20	1	0.138	×	×
ln_yp_over_y = 0.200	1.23	1	0.268	×	×
Joint wealth (4 restrictions)	10.03	4	0.040	✓	×
Joint all (6 restrictions)	29.10	6	<0.001	✓	✓

1.8.3 7.3 The reconciliation with the headline paper

The single rejection at 1 per cent is the housing-wealth m.p.c. $\gamma_{\text{HA}} = 0.0488$. Yet the headline paper reports $\gamma_{\text{HA}} = 0.049$ as the *implied* γ from Spec 6 — essentially equal to Williams' value.

The reconciliation runs through the ECM identity. Spec 6's OLS coefficient on ha_y is 0.0088; the implied $\gamma = 0.0088 / 0.180 = 0.049$ in the headline paper. Williams' implied OLS at his λ is $0.0488 \times 0.286 = 0.0140$. Our OLS is therefore 37 per cent below Williams' implied OLS, and our $|\lambda|$ is also 37 per cent below Williams'; the two ratios cancel in the implied γ comparison but not in the Wald test of the OLS restriction.

The two tests answer different questions. The implied- γ comparison asks "does our structural γ profile match Williams' Table 1?" — the answer for housing wealth is yes, almost exactly. The Wald restriction test asks "is the joint hypothesis ($\gamma_{\text{HA}} = 0.0488$ and λ as estimated) consistent with our OLS coefficients?" — and the answer is no for housing wealth because the OLS coefficient is too small relative to what Williams' γ at our λ would imply.

For the four wealth coefficients jointly, the joint Wald test rejects at 5 per cent ($\chi^2(4) = 10.03$, $p = 0.040$). For the full calibration set including the affordability and PI terms, the joint test rejects strongly ($\chi^2(6) = 29.10$, $p < 0.001$). Williams' calibrations are not consistent with a single set of OLS estimates under the Newey–West uncertainty bands.

1.9 8. Discussion and conclusion

1.9.1 8.1 What the multi-equation analysis adds

The headline single-equation paper finds that Spec 6 reproduces Williams' structural γ profile to a useful approximation, and reads the residual divergence as a consequence of the single-equation framing rather than of sample length, knot count, or sign-prior structure. The multi-equation analysis in this companion paper reaches a complementary conclusion *from a different direction*:

- The cross-equation parameter restrictions Williams' framework treats as the structural identification are only weakly supported on contemporary Australian data. One institutional event (1986 deregulation) carries the four-equation joint identification.

- Williams' Table 1 calibrations are formally rejected as a system of restrictions on the consumption equation, even though the implied γ_{HA} matches Williams almost exactly.
- The proxy HEW equation does not deliver an additional identifying signal; its residuals correlate 0.83 with the mortgage equation, so the four-equation system effectively has three independent blocks under the proxy construction.

1.9.2 8.2 Where this leaves the LIVES program for Australia

Two reasonable readings of the multi-equation evidence.

Reading A. The four-equation LIVES system, as Williams formulated it, is genuinely not the right model for contemporary Australia. The 1986 deregulation is a single institutional shock; the apparently rich array of post-1990 events (1992 banking distress, 1998 NBFI expansion, 2007 GFC, 2014/2017 APRA, 2020 COVID, 2021 APRA) does not produce sign-consistent loadings across four blocks of household behaviour. The LIVES framework's claim of a common credit-conditions factor is empirically thin once joint sign-survival is required.

Reading B. The four-equation system *is* the right model, but our proxy HEW equation is the binding constraint. A properly constructed HEW series — with the dwelling-investment subtraction Williams' definition includes — might restore additional knot survival in the joint test, and the cross-equation parameter restrictions might then become more discriminating. The 0.83 residual correlation between mortgage and HEW is consistent with this reading: under a proper HEW the residuals would have a different signal content.

Choosing between these readings requires the dwelling-investment data. Sourcing the ABS National Accounts gross fixed capital formation on residential dwellings (Cat 5206.0 Table 5) and constructing a real per-capita series anchored to the same nominal income denominator as our master is the obvious next-step data investment. Until then the multi-equation evidence above is the binding empirical constraint.

1.9.3 8.3 What does *not* change

The headline single-equation finding — that Spec 6 reproduces Williams' structural γ profile to a useful approximation, with $\gamma_{HA} = 0.049$ matching Williams almost exactly — stands. The Phase A and Phase B multi-equation evidence does not overturn it. What it adds is a more precise account of *what kind of statement* the headline finding is: an implied- γ match, conditional on a 35 per cent shortfall in both the OLS coefficient and $|\lambda|$, which cancels under the ECM identity but does not survive a joint Wald restriction. The two statements are simultaneously true and the distinction is methodological rather than empirical.

1.9.4 8.4 Path forward to full FIML

The four-equation SUR with joint-survival CCI is the natural pre- FIML configuration. The remaining methodological gap — imposing parameter equality across equations rather than only sign equality — is the Phase B item B1 in the multi-equation plan. Custom likelihood code with shared ω across the consumption and HP equations is a 2–4 week implementation; the present paper provides the data and SUR infrastructure that the FIML build would inherit.

Given the Phase A finding that the joint sign-survival CCI is a single-knot series, the FIML build would not have many cross-equation parameters to share initially. The case for committing to the months-of-work FIML is therefore weaker than it appeared before Phase A: the structural identification machinery operates on a thin empirical basis. Expanding the

candidate knot set (e.g. with additional Australian institutional events such as specific RBA policy episodes) is a logically prior step.

1.10 References

(Shared bibliography with the headline single-equation paper; see `Australia/docs/wp_draft.md` §References. Additional citations specific to this paper:)

- Duca, J. V., & Muellbauer, J. (2013). Tobin LIVES: Integrating evolving credit market architecture into flow-of-funds based macro models. *European Central Bank Working Paper* 1581.
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 - Williams, D. M. (2010). Consumption, wealth and credit liberalisation in Australia. *Oxford Economics Series Working Paper* 492.
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1.11 Appendix A: Implementation

Phase 1 SUR: `LIVES/R/lives_sur_2eq.R`.

Phase 3 joint sign-survival and three-equation SUR: `LIVES/R/joint_cci_identification.R`, `LIVES/R/lives_sur_3eq.R`.

Phase A four-equation SUR with HEW + ζ -normalisation variants: `LIVES/R/lives_sur_4eq.R`.

Phase B Williams calibration Wald tests: `LIVES/R/williams_calibration_test.R`.

Data preparation: `LIVES/R/lives_data_prep.R`. Iterated maximal-GETS CCI fitting in `Australia/R/australia_estimation.R` function `fit_consumption_with_williams_cci()`.

1.12 Appendix B: Outputs

- `lives_joint_cci_survival.csv`
- `lives_sur_2eq_coefs.csv`, `lives_sur_2eq_compare.csv`, `lives_sur_2eq_resid_corr.csv`
- `lives_sur_3eq_coefs.csv`, `lives_phase3_comparison.csv`
- `lives_sur_4eq_coefs.csv`, `lives_sur_4eq_residcorr.csv`, `lives_phase_a_summary.csv`
- `williams_calibration_wald.csv`